CHAPTER 01

The Universe & Our Solar System

Look up at the night sky. Every dot of light is a clue about a universe so huge it's hard to imagine. In this chapter you'll find out where Earth sits in space, how that idea changed over time, and what else is flying around out there.

📚 Georgia Standard S6E1

WHAT YOU'LL LEARN

- **01** How Our View Changed
- **02** Our Place in the Universe
- **03** Meet the Planets
- **04** Comparing the Planets
- **05** Gravity & Inertia
- **06** Comets, Asteroids & Meteoroids

1.1 How Our View of Space Changed

A **model** is a simpler way to show something that is hard to see all at once. For thousands of years, people built models of the sky using only their eyes. As they invented better tools, like the telescope, they collected new evidence — and their models **changed**. This is exactly how science is supposed to work: when better information shows up, scientists update their ideas.

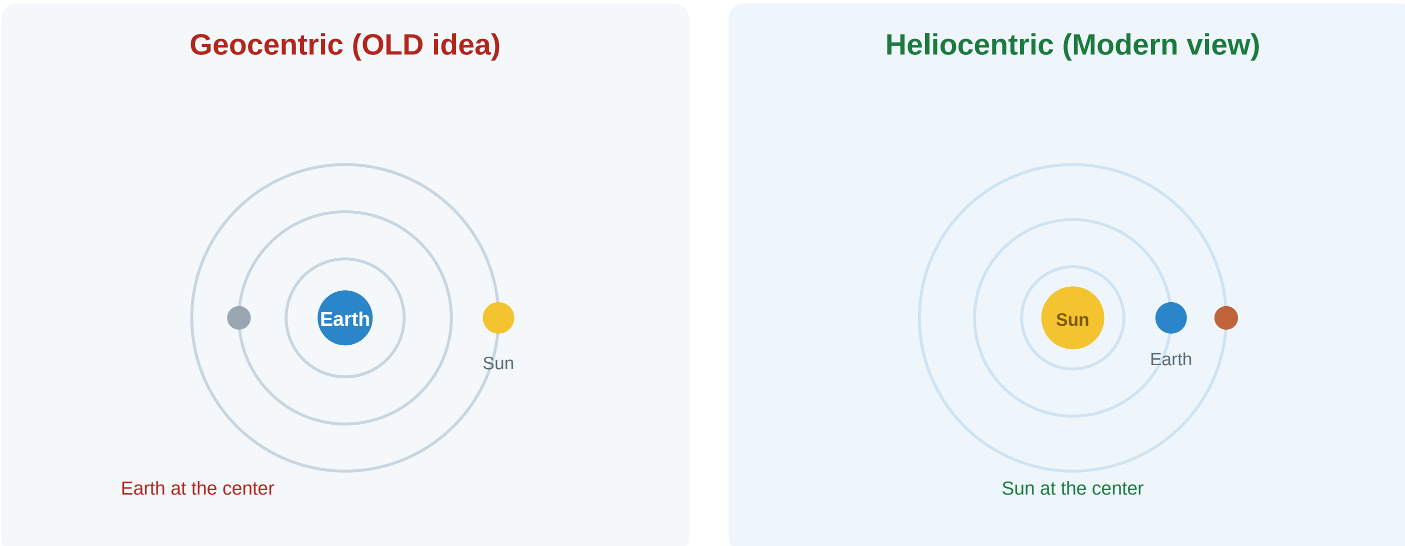


Figure 1.1 — The geocentric model put Earth in the middle. The heliocentric model, supported by better evidence, puts the Sun in the middle.

From "Earth in the Middle" to "Sun in the Middle"

Long ago, most people believed in the **geocentric model** ("geo" means Earth). They thought Earth sat still in the center while the Sun, Moon, and stars circled around it. It made sense — it really does *look* like the Sun moves across the sky each day.

Later, a scientist named **Nicolaus Copernicus** suggested the **heliocentric model** ("helio" means Sun): the Sun is in the center, and Earth and the other planets orbit around it. When **Galileo** pointed his telescope at the sky, he saw moons orbiting Jupiter — proof that not everything circles Earth. The new evidence won.

Big Idea: Science Changes

Scientific ideas are not "wrong" forever or "right" forever. They are the **best explanation we have so far**. When new evidence appears, scientists are allowed — and expected — to change the model. That's a strength, not a weakness.

Quick Check 1.1

1. What is the difference between a geocentric and a heliocentric model?
2. What tool helped scientists collect the evidence that changed their model of the solar system?

1.2 Our Place in the Universe

Earth can feel like the whole world, but it's a tiny part of something enormous. To understand where we are, scientists "zoom out" step by step.

Where Are We? Zooming Out

Figure 1.2 — Zooming out from Earth to the entire universe. Each step is much, much bigger than the one before.

Level	What It Is	Helpful Comparison
Earth	The planet we live on	One house
Solar System	The Sun plus everything orbiting it (8 planets, moons, asteroids, comets)	Your whole neighborhood
Milky Way Galaxy	A huge group of billions of stars, including our Sun	Your entire city
The Universe	All galaxies, stars, and space that exists	The whole planet Earth

Our Sun is just **one star** among hundreds of billions in the Milky Way. And the Milky Way is just **one galaxy** among billions in the universe. We live on a small planet, near the edge of one arm of one galaxy.

? Quick Check 1.2

1. Put these in order from smallest to largest: galaxy, solar system, universe, Earth.
2. What galaxy is our solar system part of?

1.3 Meet the Planets

Eight planets orbit our Sun. The four closest are small and rocky. The four farthest are giant balls of gas and ice. A fun way to remember the order is: **M**y **V**ery **E**xcellent **M**other **J**ust **S**erved **U**s **N**achos.

Planet	Type	What Makes It Special
Mercury	Rocky	Closest to the Sun; smallest planet; no atmosphere to speak of
Venus	Rocky	Hottest planet; thick poisonous atmosphere traps heat
Earth	Rocky	The only planet known to support life; liquid water on the surface
Mars	Rocky	The "Red Planet"; has the largest volcano in the solar system
Jupiter	Gas Giant	Largest planet; has a giant storm called the Great Red Spot
Saturn	Gas Giant	Famous for its bright, wide rings made of ice and rock
Uranus	Ice Giant	Spins on its side; pale blue-green color
Neptune	Ice Giant	Farthest planet; very cold, with the fastest winds

! **Common Mix-Up**

Pluto is not a planet anymore. In 2006, scientists created clear rules for what counts as a planet. Pluto didn't meet all of them, so it's now called a *dwarf planet*. This is another example of a model changing when scientists agreed on better definitions.

1.4 Comparing the Planets

When we line up data about the planets, clear patterns appear. Three things matter most: how **big** a planet is, how **far** it is from the Sun, and whether it could **support life**.

Planet	Size vs. Earth	Distance from Sun	Could It Support Life?
Mercury	Much smaller	Closest	No — too hot in day, freezing at night, no air
Venus	About the same	2nd	No — hot enough to melt lead; crushing atmosphere
Earth	Earth (the standard)	3rd	Yes — water, breathable air, the right temperature
Mars	About half	4th	Not now — too cold and thin air, but scientists study it closely
Jupiter	Much larger	5th	No — no solid surface; made of gas
Saturn	Much larger	6th	No — gas planet, extremely cold
Uranus	Larger	7th	No — freezing ice giant
Neptune	Larger	8th	No — coldest and most distant

The "Just Right" Zone

Earth sits in a special spot scientists call the **habitable zone** — not too close to the Sun (too hot) and not too far (too cold). This "just right" distance lets Earth have **liquid water**, which every living thing we know of needs. Move Earth much closer or farther and life as we know it couldn't exist.

Quick Check 1.4

1. Name two reasons Earth can support life that Mars cannot.
2. Why are the gas giants unable to support life as we know it?

1.5 Why Planets Orbit: Gravity & Inertia

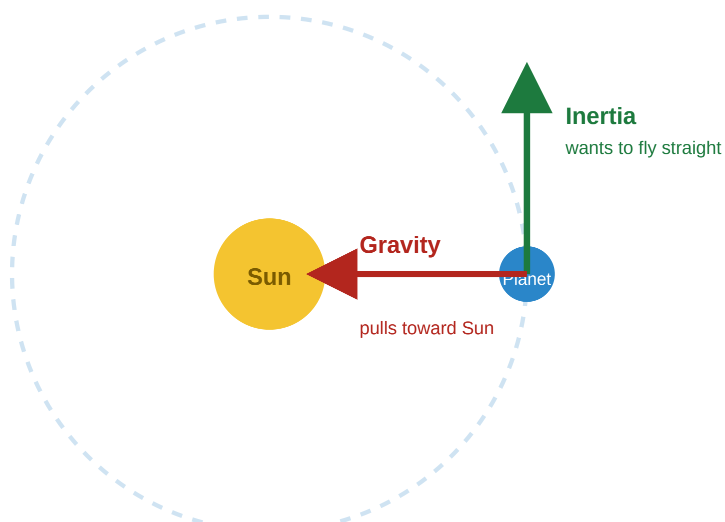
Why don't the planets just fly off into space, or crash into the Sun? The answer is a tug-of-war between two things: **gravity** and **inertia**.

Gravity — the pull

Gravity is a force that pulls objects toward each other. The more **mass** an object has, the stronger its pull. The Sun is gigantic, so its gravity reaches across the whole solar system and pulls the planets toward it.

Inertia — the "keep going straight"

Inertia is the tendency of a moving object to keep moving in a straight line. A planet is zooming through space, and inertia "wants" to send it flying off in a straight path.



The two forces combine to make a curved orbit instead of a straight line or a crash.

Figure 1.3 — Gravity pulls inward while inertia pushes straight ahead. Together they bend the path into a near-circle: an orbit.

Picture This

Imagine swinging a ball on a string in a circle. Your hand pulling inward is like **gravity**. The ball "wanting" to fly straight off is like **inertia**. As long as both work together, the ball keeps circling. Let go (no gravity) and it flies straight off — just like a planet would.

Quick Check 1.5

1. What would happen to Earth's path if the Sun's gravity suddenly disappeared?
2. Which has a stronger pull of gravity: the Sun or a small moon? Why?

1.6 Comets, Asteroids & Meteoroids

Planets aren't the only things orbiting the Sun. Smaller chunks of rock and ice are out there too. They have different names depending on what they're made of and where they are.

Figure 1.4 — Three kinds of small space objects. Each forms in a different way and looks different.

Object	Made Of	Where It's Found	Key Feature
Comet	Ice, dust, and rock ("dirty snowball")	Far edges of the solar system	Grows a glowing tail as it nears the Sun and the ice melts
Asteroid	Rock and metal	Mostly the asteroid belt (between Mars and Jupiter)	Rocky leftover chunks from the early solar system
Meteoroid	Small piece of rock or metal	Anywhere; often broken off asteroids or comets	Becomes a "shooting star" if it burns up in Earth's atmosphere

Don't Mix These Up

The same little rock can have three names! A **meteoroid** is the rock floating in space. If it streaks through our atmosphere and glows, we call that streak a **meteor** (a "shooting star"). If a piece survives and lands on the ground, it's a **meteorite**. Memory trick: **"-ite" hit** the ground.

Asteroid Belt: a ring-shaped region between Mars and Jupiter where most asteroids orbit the Sun. Think of it as the dividing line between the small rocky planets and the giant planets.

Quick Check 1.6

1. What gives a comet its tail, and when does the tail appear?
2. What is the difference between a meteoroid, a meteor, and a meteorite?
3. Where is the asteroid belt located?

Chapter 1 Wrap-Up

★ Key Takeaways

- **Models change.** We went from a geocentric (Earth-centered) to a heliocentric (Sun-centered) model as evidence improved.
- **We're small.** Earth → Solar System → Milky Way Galaxy → Universe, each far bigger than the last.
- **Eight planets** orbit the Sun: four rocky, four gas/ice giants. Only Earth is known to support life.
- **Gravity + inertia** work together to keep planets in their orbits.
- **Comets, asteroids, and meteoroids** are smaller objects, each made of different stuff and found in different places.